

CAMPUS LENZ

Complete Project Specification

CATEGORY	College Discovery + Student Experience + Social Networking
PRODUCT TYPE	College ecosystem platform
PRIMARY USERS	Student • Alumni • Institution • Admin
CORE LAYERS	Discovery • Community • Trust

A unified specification covering product vision, users, discovery, community, trust, data architecture, technical foundation, and development.

1. Project Identity

Project Name: Campus Lenz\ Category: College Discovery + Student Experience + Social Networking Platform

Core Idea

Campus Lenz is a trusted college-experience platform where students
and alumni can discover colleges, understand real student experiences,
compare institutions, connect with alumni, participate in communities,
and communicate with other campus users.

Campus Lenz combines:

- Official college information
 - Student experiences
 - Alumni experiences
 - Structured reviews
 - Ratings
 - College comparison
 - Social posts
 - Communities
 - Alumni networking
 - Messaging
 - Institution information
 - Moderation and trust mechanisms
-

2. Problem Statement

College information is scattered across college websites, Google, YouTube, Instagram, Reddit, friends, seniors, alumni, and review platforms.

This information can be:

- Incomplete
- Promotional
- Unstructured
- Difficult to compare
- Difficult to verify
- Scattered across multiple platforms

Students often need answers to questions such as:

- How is the hostel?
- Are placements actually good?
- How is the teaching?
- What is student life like?
- What do current students think?
- Can I talk to an alumnus?

Campus Lenz brings college information, real experiences, discovery, comparison, community, and alumni networking into one platform.

3. Product Vision

Campus Lenz is a college ecosystem platform, not simply a college-review website. It combines three major layers:

Layer 1 --- Discovery

- Colleges
- College Profiles
- Reviews
- Comparison
- Rankings / Discovery
- Saved Colleges

Layer 2 --- Community

- Social Feed
- Posts
- Communities
- Alumni Network
- Events
- Questions
- Mentoring
- Messaging

Layer 3 --- Trust

- Authentication
- Verification
- Moderation
- Reports
- Anonymous Experience
- Institution Controls
- Admin Review

These layers should operate as one connected product.

Campus Lenz has four primary account types.

4.1 Student

Students are the primary consumers and contributors.

Capabilities

- Discover colleges
- Read reviews
- Write reviews
- Create posts
- Provide college feedback
- Compare colleges
- Save colleges
- Join communities
- Find alumni
- Follow users
- Message users

- Participate in campus discussions

Students do not require strong verification merely to browse and use the platform.

4.2 Alumni

Alumni are both contributors and networking participants.

Capabilities

- Alumni profile
- Share college experiences
- Write alumni reviews
- Create posts
- Join communities
- Create alumni communities
- Answer student questions
- Mentor students
- Network with alumni
- Message users
- Participate in group chats

Alumni Verification

Alumni verification is separate from login authentication.

Possible verification methods:

- College email
- Alumni / student ID
- Graduation proof
- Other institution proof

Verification lifecycle:

Pending → Admin Review → Approved / Rejected

4.3 Institution

An institution represents the college or organization itself.

Capabilities

- Official institution profile
- College information
- Courses
- Departments
- Fees
- Placements
- Facilities
- Events
- Announcements
- Official posts
- Review responses
- Institution analytics

Institution Verification

Institution verification can use:

- Official email
- Institution details
- Proof / authorization
- Admin review

Institution Restrictions

Institutions cannot:

- Edit student reviews
- Delete student reviews
- Change student ratings
- Suppress legitimate criticism

Institutions can:

- Read reviews
 - Reply to reviews
 - Publish official information
 - Publish announcements
 - Publish events
 - Manage official content
-

4.4 Admin

Admin manages the platform's trust, safety, and administration layer.

Capabilities

- User management
 - Alumni verification
 - Institution verification
 - Reports
 - Content moderation
 - Community management
 - College data management
 - Platform settings
 - Suspicious activity review
 - Audit history
-

Campus Lenz has five major user-facing areas:

Area	Purpose
Home	Social / campus feed
Explore	College discovery and information
Create	Reviews, feedback, and social posts
Messages	Direct and community communication
Profile	Identity, activity, saved content, and settings

Home is the social layer of Campus Lenz.

Users can discover posts about:

- Colleges
- Student life

- Campus experiences
- Questions
- Events
- Alumni discussions
- Academic experiences
- Hostel experiences
- Placements
- General campus discussions

Example

"How is hostel food at XYZ College?"

A current student can respond, an alumnus can provide experience, and other users can continue the discussion.

This creates a campus knowledge network.

Explore is the primary discovery engine.

Users can search and filter colleges using:

- College name
- Location
- State
- College type
- Course
- Department
- Fees
- Placement information
- Rating
- Review count

The discovery process leads to the dedicated college profile.

Every college has a dedicated profile.

Basic Information

- College name
- Location
- College type
- Established information
- Contact information

Academic Information

- Courses
- Departments
- Programs

Financial Information

- Fees
- Available fee information

Facilities

- Hostel
- Library
- Labs
- Campus facilities
- Infrastructure

Placement Information

- Placement information
- Available statistics
- Student / alumni experiences

Student Experience

A dedicated section representing what students and alumni actually say about the institution.

Community Content

- Reviews
- Posts
- Feedback
- Events
- Alumni activity

User Actions

- Save
- Compare
- Share
- Write a review

Reviews are different from ordinary social posts.

A review is structured specifically to evaluate a college experience.

Required Terminology

Reviewer Type: Student or Alumni

Do not automatically display:

- Verified Student
- Verified Alumni

unless actual verification status is intentionally being displayed.

Review Dimensions

Potential dimensions include:

- Academics
- Faculty
- Placements
- Infrastructure
- Hostel

- Campus Life
- Value for Money
- Student Experience

Review Content

A review can contain:

- Overall rating
- Category ratings
- Review title
- Experience
- Pros
- Cons
- Advice
- Recommendation
- Reviewer type
- Course
- Department
- Batch

Campus Lenz must not fabricate information.

When information is unavailable:

Information not available yet.

When there is insufficient review data:

Not enough data.

Ratings should come from actual submitted reviews.

The platform must distinguish between available evidence and unavailable information.

Campus Lenz separates official information from community experience.

Official Information

Provided by the institution.

Example:

"Our college offers 42 undergraduate programs."

Community Experience

Provided by students and alumni.

Example:

"The placement department was helpful during my final year."

Both can appear on the same college profile, but the source and nature of the information remain distinct.

Users can compare up to three colleges.

Comparison Dimensions

Category	College A	College B	College C	-----	-----	-----	-----	Academics	Data	Data	Data
Faculty	Data	Data	Data	Placements	Data	Data	Data	Infrastructure	Data	Data	Data
								Hostel	Data	Data	Data

Campus Life Data Data Data Value for Money Data Data Data Student Experience Data Data Data
Campus Lenz should present the underlying information rather than automatically declaring a winner.
The user makes the final decision.

Campus Lenz can provide ranking or discovery views when the methodology is transparent.

Potential factors:

- Review volume
- Category ratings
- Student experience
- Placement information
- Academic information
- Location
- Fees
- Other documented criteria

Ranking Principle

Do not show unexplained scores such as:

"AI Score: 97.4"

unless the calculation methodology and underlying data are clearly defined.

Ranking methodology must be visible to users.

Posts are different from structured reviews.

Examples:

"Anyone from PSG Tech here?"

"How is MCA placement this year?"

"Looking for alumni from Computer Science."

"My experience with hostel facilities."

Posts create the community layer of Campus Lenz.

A post can contain:

- Text
 - Optional image
 - College / topic association
 - Visibility
 - Likes
 - Comments
 - Shares
 - Saves
 - Reports
 - Hide action
-

Because Campus Lenz is a social platform, moderation is essential.

General content lifecycle:

Create → Moderation → Classification → Publication

Normal

Publish normally.

Sensitive but Legitimate

Allow the content, with potentially limited or anonymous public display.

Example:

Anonymous Student

or

Anonymous Alumni

Harmful

Potential examples:

- Threats
- Hate
- Doxxing
- Targeted harassment
- Sexual abuse / explicit content
- Spam / scams
- Dangerous or abusive content

Possible actions:

- Hold
- Restrict
- Remove
- Report
- Human review

AI should not be the only decision-maker.

Campus Lenz must protect legitimate criticism.

Examples of legitimate criticism:

"The hostel food was poor."

"The college charged too much for hostel facilities."

"The teaching quality in my department was poor."

Negative feedback should not automatically be removed because it is negative.

Moderation should distinguish criticism from genuinely harmful content.

Campus Lenz can provide pseudonymous or anonymous public display for sensitive but legitimate content.

Public View

Anonymous Student

"I had a difficult experience with..."

Private Platform Record

The platform can retain:

- Author ID
- College ID
- Moderation status
- Audit information

- Submission information

This creates a balance between:

Student Privacy + Platform Accountability

Alumni are not simply reviewers. They are part of the networking layer.

Students can discover alumni based on:

- College
- Course
- Graduation year
- Professional information where available
- Community participation
- Career
- Interests

Potential student questions:

- How were placements?
 - How was MCA at this college?
 - What should I prepare before joining?
 - What skills helped you after graduation?
 - How can I prepare for internships?
-

Verified alumni can create alumni communities.

Communities can be based on:

- College
- Department
- Course
- Graduation batch
- Career
- Interest
- Mentoring

Example:

SNS Alumni --- MCA

Possible discussion areas:

- Career opportunities
- College experiences
- Higher studies
- Jobs
- Mentoring
- Industry guidance

Community Features

- Posts
- Members
- Events
- Information
- Community chat
- Join requests
- Member management

Public communities can be discoverable.

Private communities can require admin approval for membership.

Campus Lenz includes direct communication.

Direct Messaging

Potential relationships:

- Student → Alumni
- Student → Student
- Alumni → Alumni
- Institution → User where appropriate

Message Requests

Unconnected users should not automatically enter another user's inbox.

Recommended model:

Message Request → Accept → Conversation

Chat Features

- Text
- Images
- Replies
- Reactions
- Report
- Block

Messaging Areas

- Direct Messages
- Message Requests
- Group Chats
- Community Chats

Users can save colleges they are researching.

Example:

A student researching MCA can save:

- College A
- College B
- College C

Saved colleges create a personal college research space and can later be used with comparison.

Saved content may also include:

- Saved posts
- Saved communities where useful

Institutions have their own dashboard.

Official Content Management

Institutions can manage:

- Courses
- Departments
- Fees
- Facilities
- Contact details
- Announcements
- Events
- Official posts

Review Response

Example:

Student review:

"Hostel facilities need improvement."

Institution response:

"We have started renovation work..."

The original student review remains intact.

Institutions must not be able to manipulate community information.

They cannot:

- Edit student reviews
- Change student ratings
- Delete criticism
- Alter student posts
- Suppress legitimate negative experiences

They can respond and provide their official perspective.

Verification is separate from authentication.

Authentication

Answers:

"Who is logging in?"

Verification

Answers:

"Can this person's claimed identity or institutional relationship be trusted?"

Alumni

Evidence may establish a relationship with the institution.

Institution

Evidence may establish that the account represents the institution.

Verification Lifecycle

Pending → Admin Review → Approved / Rejected

Campus Lenz has three separate trust systems.

Authentication

Determines whether an account exists and can log in.

Verification

Determines whether an account's claimed relationship to an institution has been verified.

Moderation

Determines whether user-generated content is acceptable under platform rules.

These systems must remain separate.

Admin operates the platform management layer.

Users

- Students
- Alumni
- Institutions

Verification

- Pending
- Approved
- Rejected

Reports

- Reported posts
- Reported reviews
- Reported users
- Reported messages / communities where applicable

Moderation

- Pending content
- Moderation results
- Actions taken
- Audit history

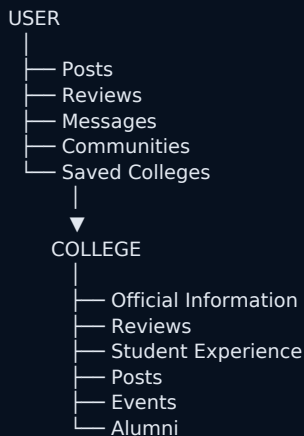
Communities

- Community management
- Reports
- Moderation

College Data

- College records
 - Academic information
 - Institutional information
 - Data corrections
-

The central relationship of Campus Lenz is:



Central Identifier

`college_id`

The college becomes a connection point between:

- Students
- Alumni
- Reviews
- Posts
- Communities
- Events
- Institution accounts
- Comparisons
- Saved colleges

This relationship should be reflected consistently in the database architecture.

The conceptual data model should include entities for:

User
Student Profile
Alumni Profile
Institution Profile
Admin
College
Course
Department
Review
Review Dimension
Post
Comment
Reaction
Saved College
Saved Post
Community
Community Member
Community Join Request
Event

Message
Message Request
Conversation
Verification Request
Report
Moderation Action
Notification
Audit Log

The exact schema should be derived from the existing implementation before new tables are created.

The existing Campus Lenz implementation already provides a Phase 1--2 foundation around:

- React / Next.js
- TypeScript
- Supabase / PostgreSQL
- Authentication
- Registration
- Onboarding
- College database
- Reviews
- Ratings
- Comparison
- Saved colleges
- RLS
- Local authentication fallback

Existing routes include:

/login
/register
/explore
/colleges/[slug]
/reviews
/reviews/submit
/compare
/dashboard
/profile

Development Rule

The existing working implementation is the baseline.

Do not:

- Break existing functionality
- Rewrite working functionality unnecessarily
- Remove existing database logic without explicit approval
- Make uncontrolled architectural changes

Before modifying code:

1. Inspect the current implementation. 2. Identify affected files. 3. Explain the proposed change. 4. Preserve unrelated functionality. 5. Implement only the requested scope. 6. Run lint / type / build checks. 7. Test navigation and user flow. 8. Report exact results.

Campus Lenz follows a dark glassmorphic mobile-first design.

Color System

Background #07111F
Surface #112238
Surface Variant #162D4A

Mint #38E6A5
Mint Light #70F3C1

Blue	#5B8CFF
Amber	#FBBF24
Violet	#A78BFA
Text	#F8FAFC
Muted	#94A3B8

Design Characteristics

- Dark background
- Glass-like surfaces
- Rounded cards
- Clear spacing
- Strong visual hierarchy
- Mobile-first layouts
- Modern social-app patterns
- Consistent visual language

Different screens should not randomly use unrelated design styles.

1. Authentication $\neq$ Verification 2. Verification $\neq$ Moderation 3. Negative feedback $\neq$ Abuse 4. Institution $\neq$ Faculty 5. Alumni verification is separate from login 6. Anonymous public display can retain private author records 7. Institutions can respond to reviews but cannot modify user reviews 8. Communities are a separate social layer 9. Messaging should protect users from unsolicited spam through message requests 10. Ranking methodology must be transparent 11. Do not invent data 12. Do not break existing working features 13. Keep the application flow connected 14. Mobile-first 15. Build in controlled phases

Campus Lenz should always distinguish:

Verified Relationship

"This account has been verified as connected to this institution."
from:

User Opinion

"This is what the user says about their experience."
Verification does not automatically make every statement factual.
Similarly:

Official Information

Information published by an institution.

Community Information

Information submitted by students or alumni.
These sources should remain distinguishable throughout the product.

The central product experience should connect discovery and community:

```

DISCOVER COLLEGE
↓
VIEW COLLEGE
↓
READ STUDENT / ALUMNI REVIEWS
↓
COMPARE COLLEGES
↓
SAVE COLLEGE

```

↓
FIND ALUMNI
↓
MESSAGE / ASK QUESTIONS
↓
PARTICIPATE IN COMMUNITY
↓
RETURN TO COLLEGE DISCOVERY

The product should feel like one connected ecosystem rather than a collection of unrelated pages.

Phase 0 --- Existing Project Audit

Inspect:

- Existing code
- Routes
- Components
- Database schema
- Supabase configuration
- RLS
- Authentication
- Existing Phase 1--2 functionality

Output:

Current State → Target State Gap Analysis

Phase 1 --- Application Shell

- Global layout
 - Session handling
 - Protected routes
 - Public routes
 - Bottom navigation
 - Mobile-first structure
-

Phase 2 --- Authentication + Onboarding

- Student registration
 - Alumni registration
 - Institution registration
 - Role-specific onboarding
 - Verification entry points
-

Phase 3 --- Home / Social Feed

- Feed
- Posts
- Comments
- Likes
- Saves
- Shares

- Reports
 - Hide
-

Phase 4 --- Explore

- College discovery
 - Search
 - Filters
 - Reviews
 - Communities
 - Alumni
 - Saved
-

Phase 5 --- College Discovery

- Search
 - Filtering
 - Results
 - College profiles
-

Phase 6 --- College Profile

- Overview
 - Courses
 - Departments
 - Fees
 - Placements
 - Facilities
 - Student Experience
 - Ratings
 - Reviews
 - Posts
 - Feedback
 - Events
 - Save
 - Compare
 - Share
-

Phase 7 --- Reviews

- Structured review form
- Reviewer type
- Evaluation dimensions
- Ratings
- Pros
- Cons
- Advice

- Recommendation
 - Moderation
 - Publication
-

Phase 8 --- Create System

- Create post
 - College feedback
 - Write review
 - Content checks
 - Moderation outcomes
-

Phase 9 --- Communities

- Community discovery
 - Community creation
 - Members
 - Posts
 - Events
 - Information
 - Chat
 - Join requests
-

Phase 10 --- Alumni Network

- Alumni search
 - Alumni profiles
 - Follow
 - Message
 - Mentor capability
-

Phase 11 --- Messaging

- Direct messages
 - Message requests
 - Group chats
 - Community chats
 - Blocking
 - Reporting
-

Phase 12 --- Institution Platform

- Institution verification
- Official profile
- Information management
- Events

- Announcements
 - Official posts
 - Review responses
 - Analytics
-

Phase 13 --- Admin Platform

- User management
 - Verification
 - Reports
 - Moderation
 - Community management
 - College data
 - Settings
 - Audit
-

Phase 14 --- Notifications

- Likes
 - Comments
 - Follows
 - Messages
 - Message requests
 - Community activity
 - Verification status
 - Institution responses
-

Phase 15 --- Responsive + UX Testing

Test:

- Mobile
 - Tablet
 - Desktop
 - Loading states
 - Empty states
 - Error states
 - Back navigation
 - Refresh
 - Session persistence
 - Unauthorized access
-

Phase 16 --- End-to-End Testing

Test all four roles:

- Student
- Alumni

- Institution
- Admin

Test complete real-world journeys rather than only individual screens.

Phase 17 --- Build / APK

Only after the application is stable:

Feature Testing
 → Role Testing
 → RLS Testing
 → End-to-End Testing
 → Responsive Testing
 → Production Build
 → APK / Deployment

Every implementation follows:

PLAN → BUILD → TEST → FIX → NEXT

Before implementation:

1. Identify where the feature belongs in the product. 2. Inspect the current implementation. 3. Identify affected files. 4. Identify database impact. 5. Identify permission / RLS impact. 6. Explain what changes. 7. Explain what does not change.

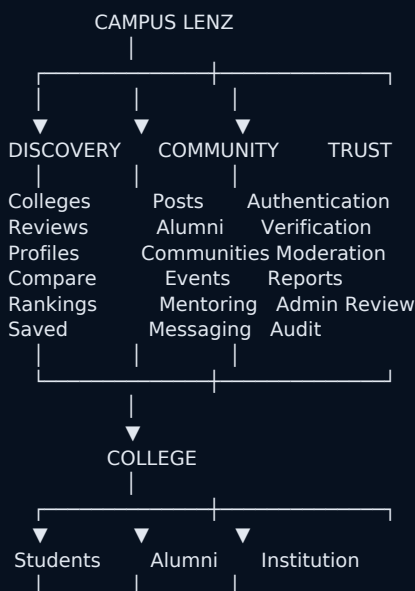
After implementation:

1. Files changed 2. Features implemented 3. Tests performed 4. Test results 5. Remaining issues

****Campus Lenz is a trusted digital campus ecosystem that combines college discovery, structured student and alumni experience data, social discussion, alumni networking, communities, and direct communication in one platform.****

The platform is built around a simple principle:

****Help students understand colleges through information, real experiences, structured comparison, and human connections --- without replacing the user's own judgment.****





Master Rule

Campus Lenz must always be treated as one connected product, not a collection of independent pages.

Every new feature must answer:

- Where does it belong?
- What existing entity does it connect to?
- What user problem does it solve?
- What data does it create/use?
- What permissions does it require?
- How does it connect to the rest of Campus Lenz?
- How will it be tested without breaking existing functionality?